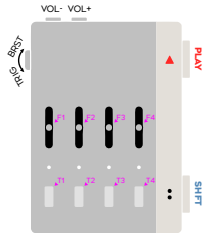


Wakes v0.5.0 - Pg.1 - Welcome



• Hey! I'm Adara. Welcome to **Wakes**, a custom firmware for **Teenage Engineering's** unreleased "SP-1" stem player. **Wakes** adapts code from **three** Eurorack modules into one device: **Plaits**, a macro-synthesizer, **Marbles**, a playful random sequencer, and **Yarns**, a signal converter. The modules were created by **Émilie Gillet**, otherwise known as **Mutable Instruments**.

• First and foremost, you should know how to turn the device on or off. Holding down **SHFT** for 1.5 seconds will show a brief animation before the device will turn on, going to its first page, **Plaits**.

• While turned on, holding **SHFT** for 3 seconds will turn the device off.

• While plugged in, **Wakes** has a standby mode.

• While on, any input while **SHFT** is held down will interrupt the shutdown behaviour, allowing you to work inside **SHFT** pages.

• Even if interrupted, holding down **SHFT** for 25 seconds will cause the face LEDs to flash quickly, notifying you that the SP-1 is about to perform an emergency shutdown in 5 seconds if you don't let **SHFT** go. Release and return to holding **SHFT** to get back to what you were doing. This is intended as an emergency backstop.

• At any point, even when off, tapping **SHFT** for a short moment will bring up a battery meter.

• While using **Wakes**, you'll be bouncing between **Plaits** to design sounds, and **Marbles** to sequence it.

• As you bounce back and forth, the face LED's will show you what's going on through a system of **glyphs**.

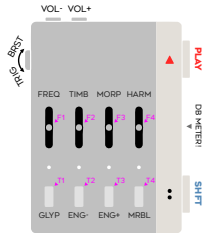
• Faders **F1-F4** have a catch-up mechanism with the actual digital values, so moving them around and swapping pages doesn't create unexpected behaviours. For clarity, follow the LED's to know where the set values reside.

• Each module has a **SHFT** page containing advanced parameters and options that open up avenues for experimenting.

• Double-tapping **SHFT** enters a deep tinkering page called **SETTINGS**. There are two **SETTINGS** pages: One for **Plaits**, one for **Marbles**.

• Legend: ○ - Lit LED ● - Unlit LED





- **Wakes'** first page is **Plaits**: A synthesizer containing a multitude of sound engines representing different sound design techniques, with parameters attached to explorative macros. A macro-synthesizer.

- **Plaits** has four main faders: **FREQ**, **TIMB**, **MORP** and **HARM**. Each of these controls several components of the selected synthesis engine.

- Plaits has a built-in lowpass-gate and amplifier, controlled by an internally-routed **decay envelope**.

- For now, you can read about what the main parameters do for each engine in commonly available Plaits guides.

- While in Plaits, the four side LED's will show a dB meter. Each LED represents 6dB of dynamic range.

- Generally speaking, **FREQ** will control the overall pitch of the sound. **TIMB** sweeps the spectral content from dark/sparse to bright/dense. **MORP** explores lateral timbral variations. **HARM** controls the frequency spread/overall balance of the spectrum.

- The rocker switch plays notes, triggering the decay envelope on the filter/amp. Press it down to send a single trigger, pull it up to send a burst.

- Engines are known by glyphs. Pressing **GLYP** will flash the glyph representing the selected engine. **ENG+** for the next one, **ENG-** to go back.

- **PLAY** engages **Marbles'** clock, which is patched by default to trigger Plaits and give it musical notes.

- Press **MRBL** to go to **Marbles**.

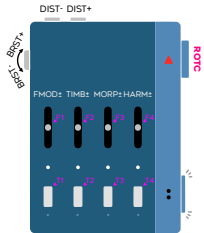
- Hold **SHIFT** to go to **Plaits SHIFT**:

- Each fader here represents a modulation depth control for incoming control signals. By default, Plaits' **decay envelope** is assigned to all of the four parameters. Use the faders to modulate Plaits' macros! Fun!
- The rocker switch selects a rhythmic subdivision for **BRST**'s retriggers.

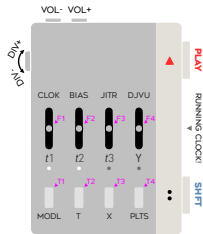
- **DIST-** and **DIST+** control the depth of a master softclipper.

- Holding down any of **T1-T4** for 2 seconds **Unpatches** Marbles modulators from that parameter.

- Holding down **PLAY** for 3 seconds **Rips Out Plaits' Cables**, returning it to its default configuration.



Wakes v0.5.0 - Pg.3 - Marbles - t Side (left)

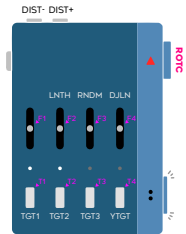


- Marbles has two sides: **t** and **X**. The glyphs will show you which side you are on: ○○○● for **t** and ●●○○ for **X**. Each side has 3 outputs: **t1-t3**, **X1-X3**. An extra output **Y** is offered. It's weird. :)
- **t** is a random gate generator. It emulates a player following a stable metronome. **CLOK** determines the BPM (default: 120). **BIAS** determines the distribution of gate signals depending on the selected model using **MODL**. **JITR** changes the player's accuracy. stable > chaotic.
- **DJVU** controls the probability of reusing played patterns. At center, it is a locked loop. Moving down adds more unlocked randomness. Up from centre, memorized notes are randomized instead. **DJLN** sets loop length.

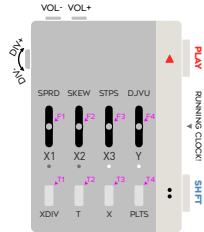
- **t2** is always a stable clock. It's a 50/50 gate, 50% on, 50% off.
- Pressing **MODL** swaps between a list of distribution models for **t1** and **t3**:
 - - coin toss: gates are evenly distributed.
 - - clusters: random pairs of clock subdivisions appear at each side.
 - - drums: Steps through drum pattern tables containing rhythms.
 - - independent coin: each side tosses its own. **t1** and **t3** can coincide.
 - - divider: **BIAS** selects from fixed subdivision pairs. 8:1 > 1:1 > 1:8.
 - - three states: a three-way choice per tick: **t1**, **t3** or neither.
- **DIV-/DIV+** divide/multiply the clock

- Press **T3** to go to the **X** side of Marbles.
- Press **PLTS** to go back to **Plaits**.
- Hold **SHFT** to go to **t SHFT**:
 - **T1-T4** select **Plaits** modulation targets for **t1-t3**, or for **Y**. The mod targets are: ●●●●-none, ○●●●-FM, ●○○●-TIMB, ●●○○-MORP, ●○○○-HARM, ●○○○-LEVL, ●○○○-TRIG. For **Y**, ●○○○ routes to **FREQ**, contextually.
 - **LNTH** determines the length of on-states for **t1** and **t3** gate signals.
 - **RNDM** determines gate length randomization for **t1** and **t3**.
- Holding down any of **T1-T4** for 2 seconds **Unpatches** any **Plaits** destinations.
- Holding **PLAY** for 3 seconds **Rips Out** all of **Marbles' Cables**.

Marbles - t SHFT



Wakes v0.5.0 - Pg.4 - Marbles - X Side (right)



- **X** is a random modulation generator. Its three outputs, **X1-X3**, are clocked by the outputs of Marbles' **t** side, and generate random control signals that can vary from Sample & Hold style steps to smooth, meandering modulations. It can modulate Plaits' parameters, or quantize it to play musical intervals, creating melodic sequences, as well as enabling many other ideas.

- The fourth output, **Y**, shows up on this page as well. It is a smooth, meandering full-range random source that's clocked at 1/8th the rate of **t2**. **Y** never obeys **DJVU**, she's wild like that.

- **SPRD** spreads the concentration of control signals around the central point determined by **BIAS**.

- **SKEW** skews the signals **X1-X3** up or down in average intensity.

- **STPS** smooths or tightens the vertical steppiness of the control signals. Below-centre, the signals have random intensities. Above-centre, it quantizes **X** to a musical scale.

- **DJVU** here is the same as on the **t** side. In fact, both sections are controlled by the same memory.

- Press **XDIV** to choose the intensity diversity between the **X** outputs: identical, bump (**X2** is strongest) or tilt (**X1** is weakest, **X3** is strongest).

- Press **T2** to go back to **t**.
- Press **T4** to go back to Plaits.

Hold **SHIFT** to go to **X SHIFT**:

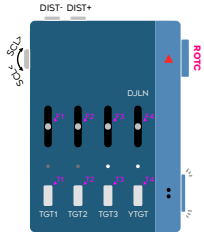
- **T1-T4** select Plaits modulation targets for **X1-X3**, or for **Y**. The mod targets are: ●●●●-none, ○●●●-FM, ●○●●-TIMB, ●●●○-MORP, ●●●○-HARM, ●○●○-FREQ, ●●○●-LEVL. FREQ is the default input for musical tuning. FM is full-range.

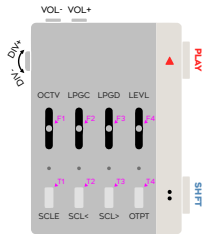
- **DJLN**, like in **t**, controls the length of the **DJVU** sequence. **DJVU** and **DJLN** are the same across the two sides.

- **SCL** select from a range of scales to quantize into, when **STPS** is above the middle point. View base available scales in the Marbles **SETTINGS** guide page.

- Holding down any of **T1-T4** for 2 seconds **Unpatches** any Plaits destinations. Hold **PLAY** to **ROTC**.

Marbles - X **SHIFT**





- Reach Plaits' **SETTINGS** page by double-tapping **SHFT** while in the Plaits module. Single-tapping **SHFT** will return you to where you were.

- **OCTV**'s default is its maximum value. When not maxed, it locks Plaits' available frequencies to 2 octaves. Choose the octave range to play by moving the fader up and down. Maximize it to go full-range again.

- **LPGC** lets you control the color of the low-pass gate, which is composed of a filter and amplifier. Lower values mix in more of the low-pass filter, making the internal decay envelope and **LEVL** sound duller in tone.

- **LPGD** lets you change the length of the decay envelope. Higher values set longer lengths, meaning triggered notes decay for longer when played. Any self-patching from the unpatched **SHFT** faders will also be longer. Shorter values sound snappier and quieter overall.

- **LEVL** changes the base offset applied to Plaits' filter and amp. Moving it around fades the sound in and out. The color of this fade is affected by **LPGC**. Setting **LEVL** to maximum will effectively disable them, allowing Plaits to drone. Some Plaits engines will react beautifully to modulating **LEVL**. It's available as a mod destination for t and X. It's the default mod destination affected by Plaits' decay envelope.

- **Wakes** includes a smart Plaits quantizer feature. It lets you noodle around with notes on a scale by moving **FREQ**, without Marbles' modulation. It is **•••••**-off by default. Activate it by pressing **SCL>**. Deactivate it by pressing **SCL<** until the glyphs show **•••••**-off again. Once activated, it first disables any Marbles routings going into **FREQ**. Then, it takes over **FREQ** and quantizes it. Combine it with **OCTV** to lock your available notes to a short range. **SCLE** flashes the currently-active quantizer scale. Scales are the same as the ones available for Marbles, with glyphs shown in Marbles' **SETTINGS** manual page.

- Plaits features an extra output, **AUX**, containing additional content from each engine. **Wakes** gives you **OTPT** on this page to select from different output configurations: **•••••**-OUT, **•••••**-AUX, **•••••**-OUT+AUX, and the **Wakes**-unique **•••••**-OUT×AUX.

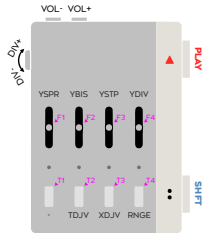
Plaits Tips

- Plaits is designed to be really breakable. Experiment with fast **BRST**, quick Marbles modulations, lots of mod targets. Just go crazy with it, and see what wilderness you end up in.

Ripping Out The Cables will take you back home.

- **FM** is a legacy Plaits input containing exp-TZFM for Plaits. It's used for atonal modulations.

- You can **TRIG** while **LEVL** is open ;)



- Reach Marbles' **SETTINGS** page by double-tapping **SHIFT** while in the Marbles module. Single-tapping **SHIFT** will return you to where you were.

- **YSPR** lets you control Y's spread, same way you do for X.

- **YBIS** lets you control Y's randomization skew towards higher or lower signal strengths.

- **YSTP** lets you control its overall steppiness. Lower values are more sinusoidal. Central values are linear, triangular. Higher values are slewed squares, up to an even sample & hold.

- **YDIV** lets you choose at which subdivision of the clock Y is clocked. This lets you accelerate it to higher values if you want.

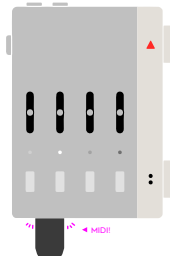
- **TDJV** and **XDJV** toggle **DJUV** for the torX sides on or off. **DJUV** is on for both by default. If you toggle it off for one of the sides, that side won't be affected by **DJUV**'s looping.

- **RNGE** sets the control signal strength range for X and Y. The options are: ○○○○ - 0-20%, ○○○● - 0-50%, ○○○○ - ±50%, ○●○○ - INTELLIGENT. INTELLIGENT is the default. It's a Wakes-specific upside of combining the two modules in a closed environment: It decides your voltage range selection for you, per-output, automatically, and with the target Plaits engine in mind. That said, if you still want to play around with unexpected results, the other modes remain available.

- There are 6 default scales Marbles offers for quantizing the X outputs. When you press **SCL<** or **SCL>** on Marbles' **SHIFT** page, their glyphs will flash on the face LEDs as you cycle through the list. Here they are: ○●○○-major, ○○○●-minor, ○○○○-pentatonic, ○○○○-pelog, ●○○○-bhairav, ●○○○-shri. If you choose one of these scales, then move **STPS** above-center, then send one of the X outputs into **FREQ**. Marbles will tell Plaits to play notes on that scale.

Marbles Tips

- You can combine multiple Marbles outputs into one Plaits destination. Using **Unpatch** disconnects only that specific output from its destination.



- Starting at v0.5.0, Wakes supports MIDI 1.0, over the SP-1's USB-C port.
- Connect your SP-1 to a MIDI host, and it should pop up as a MIDI device.
- Wakes automatically routes pitch and MIDI notes into **Plaits**, and clocks **Marbles** from the MIDI host.
- Wakes will **INTELLIGENTLY** decide control ranges and scale your CC appropriately for unipolar and bipolar parameters across the device.
- Wakes' repo includes a MIDI config file: `config/midi.ini`. Review the file for advanced information about MIDI handling.
- Fork the repository to build your own `midi.ini` and customize Wakes' MIDI handling to your needs.

PLAITS - MIDI BEHAVIOURS

- Plaits will now receive pitch, pitchbend, gate and MIDI CC information from incoming MIDI connections.
- **FREQ's** centre-detent will now expand to 10% to make it easier for you to tune Plaits to A=440.
- Plaits is a **monophonic** voice. It does not support polyphony outside of 3 engines: **Chords**, **String** and **String Machine**, and those have distinct polyphony behaviours that don't constitute traditional polyphony. This is what allows it to be a deep sound source at such a small hardware cost.
- Plaits' **MODEL** input is now exposed via MIDI CC, letting you select engines from the host. Caution - high CPU.

MARBLES - MIDI BEHAVIOURS

- Marbles will sync to external MIDI clocks until MIDI is disconnected. MIDI **START** is now the same as pressing **PLAY**.
- Marbles' **CLOK** fader now selects a ratio of the incoming MIDI clock which Marbles can play with. The ratios are: 1/4, 1/3, 1/2, 2/3, 1 (at centre), 3/2, 2, 3 or 4.
- Marbles' **DIV+** and **DIV-** rocker switch controls ratcheting of that subdivision: multiplying the beats by `○○●●-1/4`, `●○○●-1`, and `●●●○-4`.

DEFAULT MIDI CC LIST

PLAITS

3-TIMB, 9-MORP, 14-HARM, 15-FREQ, 20-FMOD±, 21-TIMB±, 22-MORP±, 23-HARM±, 24-LPGC, 25-LPGD, 26-LEVL, 102-OCTV, 105-MODEL, 64 - Sustain pedal.

MARBLES

t

27-RATE, 16-BIAS, 85-JITR, 17-LNTH, 86-RNDM,

X

28-SPRD, 29-SKEW, 30-STPS,

Y

18-YSPD, 19-YBIS, 87-YSTP, 104-YDIV

DJVU

31-DJVU, 103-DJLN